Experiment 6

Aim: Apply K-Means and Hierarchical Clustering on sample datasets

Dataset Source:

Mall Customer Segmentation Dataset
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python/data

1	CustomerI	Gender	Age	Annual Inc	Spending Score (1-100)
2	1	Male	19	15	39
3	2	Male	21	15	81
4	3	Female	20	16	6
5	4	Female	23	16	77
6	5	Female	31	17	40

Dataset Description:

The dataset contains customer information collected from a shopping mall. It is used to analyze customer behavior and perform segmentation.

Features:

- **CustomerID**: Unique identifier for each customer
- **Gender**: Male or Female
- **Age**: Age of the customer
- **Annual Income (k$)**: Annual income in thousand dollars
- **Spending Score (1–100)**: Score assigned based on customer spending behavior

Dataset Characteristics:

- Number of records: 200
- Number of features: 5
- Type: Unsupervised dataset (no target variable)

Use Case:

- Customer segmentation
- Targeted marketing
- Business strategy planning

Mathematical Formulation of the Algorithm:

K-Means Clustering

K-Means aims to partition the dataset into **K clusters** by minimizing the within-cluster variance.

$$J = \sum_{i=1}^{K} \sum_{x \in C_i} ||x - \mu_i||^2$$

Where:

- K: Number of clusters
- Ci: Set of points in cluster i
- μi: Centroid of cluster i

Hierarchical Clustering

Hierarchical clustering builds a tree-like structure (dendrogram) using a bottom-up approach.

Distance between two points:

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

Linkage methods:

- Single linkage
- Complete linkage
- Average linkage
- Ward's method

Algorithm Limitations:

K-Means Limitations:

- Requires pre-defined number of clusters (K)
- Sensitive to outliers and noise
- Assumes clusters are spherical
- Results depend on initial centroid selection

Hierarchical Clustering Limitations:

- Computationally expensive for large datasets
- Difficult to determine optimal number of clusters
- No backtracking once clusters are merged
- Sensitive to noise and outliers

Methodology / Workflow:

Steps Involved:

1. Load dataset

```
from google.colab import files
uploaded = files.upload()
```

```
Choose Files Mall_Customers.csv
Mall_Customers.csv(text/csv) - 3981 bytes, last modified: 4/6/2026 - 100% done
Saving Mall_Customers.csv to Mall_Customers.csv
```

2. Data preprocessing
   - Remove unnecessary columns (CustomerID)
   - Encode categorical data (Gender)

```
df = pd.read_csv('Mall_Customers.csv')
df.head()
```

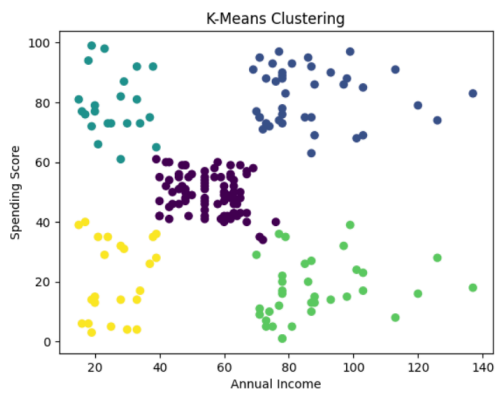
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

3. Feature selection
   - Use Annual Income and Spending Score

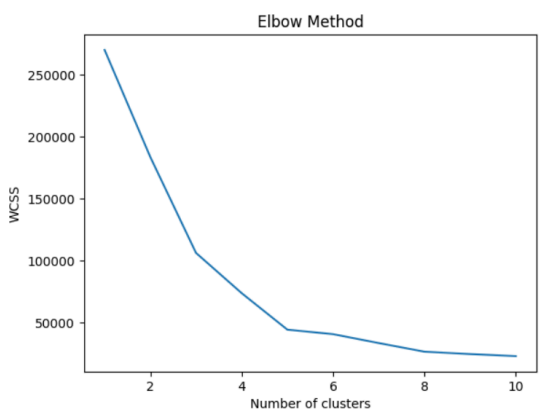
```python
# Drop CustomerID
df = df.drop('CustomerID', axis=1)

# Encode Gender
df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})

df.head()
```

	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	0	19	15	39
1	0	21	15	81
2	1	20	16	6
3	1	23	16	77
4	1	31	17	40

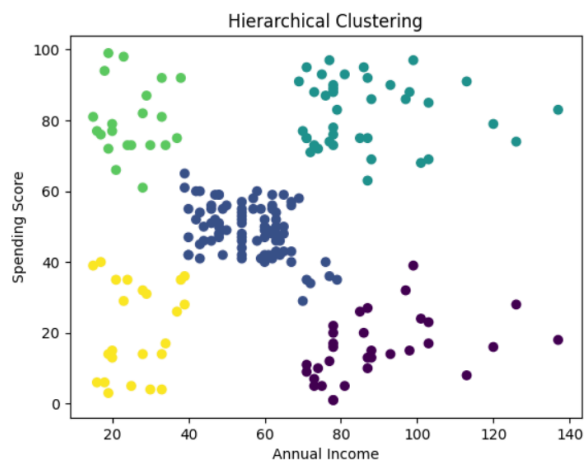
4. Apply K-Means clustering



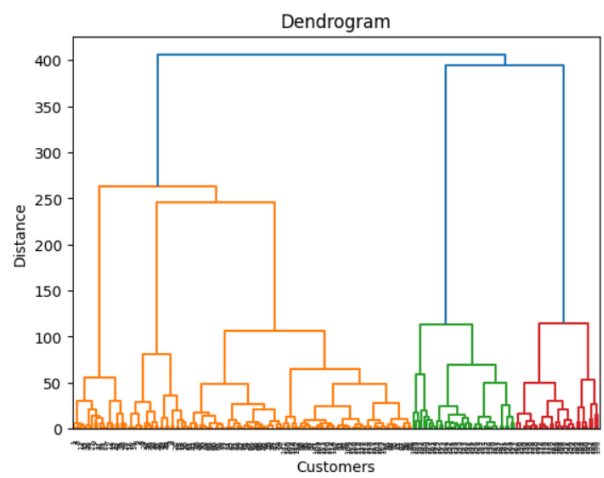
5. Determine optimal clusters using Elbow Method



6. Apply Hierarchical clustering



7. Plot dendrogram



8. Visualize clusters
9. Evaluate using Silhouette Score

```
from sklearn.metrics import silhouette_score

print("K-Means Score:", silhouette_score(X, y_kmeans)
print("Hierarchical Score:", silhouette_score(X, y_hc
```

```
K-Means Score: 0.553931997444648
Hierarchical Score: 0.5529945955148897
```

Workflow Diagram:

Dataset → Preprocessing → Feature Selection →
K-Means → Evaluation
Hierarchical → Evaluation → Visualization

Performance Analysis:

Metrics Used:

- **WCSS (Within Cluster Sum of Squares)** for K-Means
- **Silhouette Score** for both algorithms

Observations:

- Optimal number of clusters found using Elbow Method is **5**
- K-Means forms well-separated clusters
- Hierarchical clustering provides a visual understanding through dendrogram

Interpretation:

- Customers can be grouped into:
    - High income & high spending (premium customers)
    - High income & low spending
    - Low income & high spending
    - Low income & low spending

Hyperparameter Tuning

K-Means:

- Parameter tuned: `n_clusters`
- Method used: Elbow Method
- Result: Optimal K = 5

Hierarchical Clustering:

- Parameters tuned:
    - Linkage method (ward used)
    - Number of clusters

Impact:

- Proper tuning improves cluster separation and model performance

Conclusion

K-Means and Hierarchical Clustering are effective techniques for customer segmentation. These methods help businesses identify different customer groups and enable better decision-making for marketing strategies and customer engagement.

SDG Goal	How it is Implemented in the Experiment
SDG 8: Decent Work and Economic Growth	Customer segmentation using clustering helps businesses identify high-value customers and improve marketing strategies, leading to increased revenue, business growth, and potential job creation.
SDG 9: Industry, Innovation and Infrastructure	The use of machine learning algorithms like K-Means and Hierarchical Clustering demonstrates innovation in data analytics and supports the development of intelligent business systems.
SDG 12: Responsible Consumption and Production	By analyzing customer behavior, businesses can optimize production and inventory, reducing waste and promoting efficient use of resources.
SDG 17: Partnerships for the Goals	The experiment reflects collaboration between data science and business domains, encouraging partnerships for implementing scalable and data-driven solutions.